Questions

Q1.

The function f is defined by

$$f(x) = \frac{x+3}{x-4} \qquad x \in \mathbb{R}, x \neq 4$$

(a) Find ff(6)

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(2)

(b) Find f^{-1}

(3)

The function g is defined by

$$g(x) = x^2 + 5 \qquad x \in \mathbb{R}, x > 0$$

(c) Find the exact value of *a* for which

$$gf(a) = 7$$

(3)

(Total for question = 8 marks)

(**Q02 WMA13/01, Oct 2023**)

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Q2.

$$f(x) = 3 - \frac{x-2}{x+1} + \frac{5x+26}{2x^2-3x-5} \qquad x > 4$$

(a) Show that

$$f(x) = \frac{ax+b}{cx+d} \qquad x > 4$$

where *a*, *b*, *c* and *d* are integers to be found.

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(4)

(b) Hence find $f^{-1}(x)$

(2)

(c) Find the domain of f^{-1}

(2)

(Total for question = 8 marks)

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(Q03 WMA13/01, Jan 2021)

Q3.

The function f is defined by

$$f(x) = \frac{5x}{x^2 + 7x + 12} + \frac{5x}{x + 4} \qquad x > 0$$

(a) Show that $f(x) = \frac{5x}{x + 3}$

(3)

(b) Find f^{-1}

(3)

(c) (i) Find, in simplest form, $f'(x)$.

(ii) Hence, state whether f is an increasing or a decreasing function, giving a reason for your answer.

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(3)

(Total for question = 9 marks)

(Q01 WMA13/01, Oct 2021)

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Q4.

The function f is defined by

$$f(x) = \frac{5x - 3}{x - 4} \qquad x > 4$$

(a) Show, by using calculus, that f is a decreasing function.

(3)

(b) Find f^{-1}

(3)

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(c) (i) Show that $ff(x) = \frac{ax + b}{x + c}$ where a, b and c are constants to be found.

(ii) Deduce the range of ff.

(5)

(Total for question = 11 marks)

(Q06 WMA13/01, Jan 2022)

Q5.

The functions f and g are defined by

$$f(x) = 9 - x^2 \qquad x \in \mathbb{R} \quad x \geqslant 0$$

$$g(x) = \frac{3}{2x + 1} \qquad x \in \mathbb{R} \quad x \geqslant 0$$

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(a) Write down the range of f

(1)

(b) Find the value of fg (1.5)

(2)

(c) Find g^{-1}

(3)

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(Total for question = 6 marks)

(Q01 WMA13/01, Jan 2023)

Q6.

Express

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$$\frac{6x+4}{9x^2-4} - \frac{2}{3x+1}$$

as a single fraction in its simplest form.

(Total for question = 4 marks)

(Q01 WMA13/01, Specimen papers)

Q7.

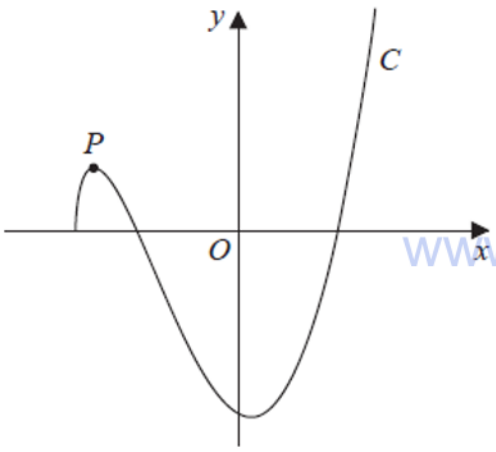


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The function f is defined by

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$$f(x) = 5(x^2 - 2)(4x + 9)^{\frac{1}{2}} \qquad x \geqslant -\frac{9}{4}$$

(a) Show that

$$f'(x) = \frac{k(5x^2 + 9x - 2)}{(4x + 9)^{\frac{1}{2}}}$$

where k is an integer to be found.

(4)

(b) Hence, find the values of x for which $f'(x) = 0$

(1)

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Figure 3 shows a sketch of the curve C with equation $y = f(x)$.

The curve has a local maximum at the point P

(c) Find the exact coordinates of P

(2)

The function g is defined by

$$g(x) = 2f(x) + 4 \qquad -\frac{9}{4} \leq x \leq 0$$

(d) Find the range of g

(3)

(Total for question = 10 marks)

(Q06 WMA13/01, June 2022)

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Q8.

The function f and the function g are defined by

$$f(x) = \frac{12}{x + 1} \qquad x > 0, x \in \mathbb{R}$$

$$g(x) = \frac{5}{2}\ln x \qquad x > 0, x \in \mathbb{R}$$

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(a) Find, in simplest form, the value of $fg(e^2)$

(2)

(b) Find f^{-1}

(3)

(c) Hence, or otherwise, find all real solutions of the equation

$$f^{-1}(x) = f(x)$$

(3)

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(Total for question = 8 marks)

(Q02 WMA13/01, Jan 2020)

Q9.

The functions f and g are defined by

$$f: x \mapsto e^x + 2 \qquad x \in \mathbb{R}$$

$$g: x \mapsto \ln x \qquad x > 0$$

(a) State the range of f.

(1)

(b) Find $fg(x)$, giving your answer in its simplest form.

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(2)

(c) Find the exact value of x for which $f(2x + 3) = 6$

(4)

(d) Find f^{-1} stating its domain.

(3)

(e) On the same axes sketch the curves with equation $y = f(x)$ and $y = f^{-1}(x)$, giving the coordinates of all the points where the curves cross the axes.

(4)

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(Total for question = 14 marks)

(Q06 WMA13/01, Specimen papers)

Q10.

The functions f and g are defined by

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$$f(x) = 5 - \frac{4}{3x+2} \qquad x \geqslant 0$$

$$g(x) = \left| 4\sin\left(\frac{x}{3} + \frac{\pi}{6}\right) \right| \qquad x \in \mathbb{R}$$

(a) Find the range of f

(2)

(b) (i) Find $f^{-1}(x)$

(ii) Write down the domain of f^{-1}

(3)

(c) Find $fg(-\pi)$

(2)

(Total for question = 7 marks)

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(Q02 WMA13/01, Oct 2022)

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